

LESSON 9-2 • HOMEWORK PACKET • SHEET 1 OF 2

From Table to Graph

Maryland Standard 6.EE.C.9 • Learning target – I can turn a table of values into points and plot them on a graph.

REMEMBER

- Each row of a table becomes an **ordered pair** (x, y) .
- To plot (x, y) , start at $(0, 0)$. Move **right** x , then **up** y .
- Points from a constant rate line up in a **straight line**.

WORKED EXAMPLE

Movie tickets cost \$8 each. Write ordered pairs for 1, 2, and 3 tickets.

Step 1 Tickets is x . Cost is y . Multiply each x by 8.

Step 2 $8 \times 1 = 8$, $8 \times 2 = 16$, $8 \times 3 = 24$

Answer $(1, 8)$, $(2, 16)$, $(3, 24)$

PRACTICE

show your work • label units

1. Write each table row as an ordered pair (x, y) .

a. $x = 4, y = 12$ _____

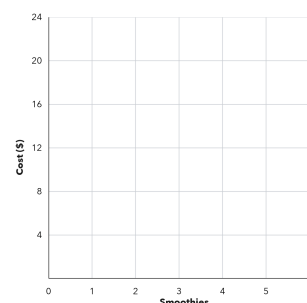
b. $x = 0, y = 0$ _____

c. $x = 6, y = 30$ _____

d. $x = 9, y = 2$ _____

2. **Smoothies.** Smoothies cost \$4 each. Complete the table. Then plot the points on the grid and connect them.

Smoothies (x)	1	2	3	5
Cost in \$ (y)				

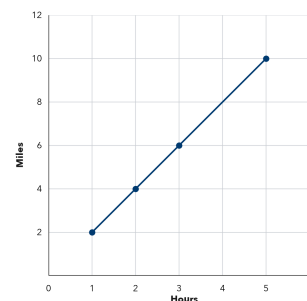


3. **Hiking.** The graph shows how far Kofi hikes.

a. Write the ordered pair for 5 hours. _____

b. What does $(3, 6)$ mean? _____

c. Miles per hour: _____



4. **Bike Rental.** A bike rents for \$6 per hour. Write the ordered pairs (hours, cost) for renting 1, 3, and 4 hours. Then tell what $(0, 0)$ means.

Ordered pairs _____

5. **Find the Error.** Teo's table shows x (hours): 2, 4 and y (dollars): 10, 20. He plotted $(10, 2)$ and $(20, 4)$. Explain his mistake. Which points should he plot?

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Show someone at home how to plot $(3, 5)$: right 3, then up 5.

LESSON 9-2 • HOMEWORK PACKET • SHEET 2 OF 2**Analyzing a Graph**

Maryland Standard 6.EE.C.9 • Learning target – I can read a relationship off a graph and describe the rate it shows.

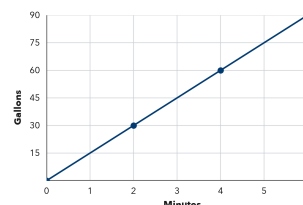
REMEMBER

- For a line through $(0, 0)$, the **unit rate** is $y \div x$. The point $(3, 12)$ means 3 of x go with 12 of y .
- With the same scales, a **steeper** line has a greater rate.

WORKED EXAMPLEA line through $(0, 0)$ passes through $(2, 10)$. Find the unit rate and y when $x = 7$.**Step 1** Unit rate: $10 \div 2 = 5$ **Step 2** $y = 5 \times 7 = 35$ **Answer 5 per unit; $y = 35$** **PRACTICE***show your work • label units*

- 1. Filling a Pool.**
- The graph shows the water in a pool.

- a.** What does $(4, 60)$ mean? _____
- b.** Gallons per minute: _____
- c.** Gallons after 9 minutes: _____



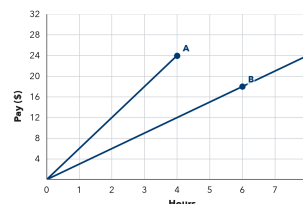
- 2.**
- Each line passes through
- $(0, 0)$
- and the given point. Find the unit rate.

- a.** $(3, 21)$ _____ **b.** $(5, 40)$ _____ **c.** $(8, 96)$ _____

- 3. Reading.**
- A line through
- $(0, 0)$
- shows that Ana reads 18 pages in 3 hours. Find her unit rate. How many pages does she read in 7 hours?

Answer _____

- 4. Two Jobs.**
- The graph shows pay for two jobs. Find each pay rate. Which line is steeper, and what does it mean?

Answer _____

- 5. Savings.**
- Jada's savings graph passes through
- $(0, 10)$
- ,
- $(1, 15)$
- , and
- $(2, 20)$
- . She says it is proportional. Is she right? What does
- $(0, 10)$
- mean?

TEST PRACTICE*choose the best answer*

- 6.** A line through the origin passes through $(5, 35)$. What is the unit rate?
- A** 5 **B** 30 **C** 7 **D** 175
- 7.** Line P passes through $(1, 4)$. Line Q passes through $(1, 9)$. Both start at $(0, 0)$. Which is true?
- A** P is steeper. **B** Q is steeper; greater rate.
- C** They have the same rate. **D** Q is steeper; smaller rate.

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____